FINC5001
Workshop 3

Presented by
Craig Mellare
University of Sydney Business School

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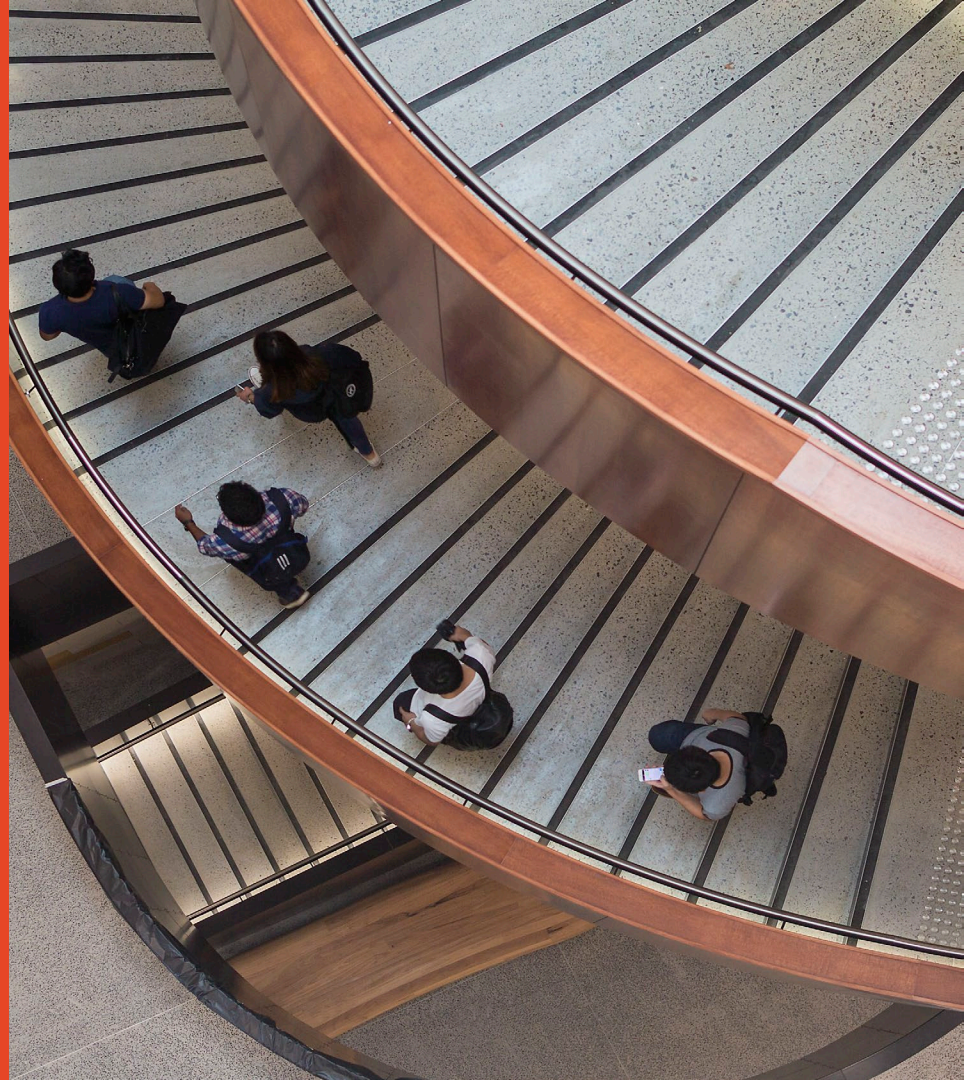

Topic One - Recap Question

CDM bank is offering deposit accounts with the following interest rates:

- *Option A: 4.95% pa compounded annually*
- *Option B: 4.9% pa compounded semi-annually*
- *Option C: 4.85% pa compounded quarterly*
- *Option D: 4.8% pa compounded monthly*

You are considering depositing $10,000.

i) Which account will give you the biggest account balance in 5 years' time?

Topic One - Recap Question

i) Calculate the effective rate of each option (the amount of money is irrelevant in deciding which option is better)

Option A: 0.0495 (this is already an effective rate)

Option B: $Effective\ rate = \left(1 + \frac{r_{nom}}{m}\right)^{mn} - 1 = \left(1 + \frac{0.049}{2}\right)^{2} - 1 = 0.0496$

Option C: $Effective\ rate = \left(1 + \frac{r_{nom}}{m}\right)^{mn} - 1 = \left(1 + \frac{0.0485}{4}\right)^{4} - 1 = 0.04939$

Option D: $Effective\ rate = \left(1 + \frac{r_{nom}}{m}\right)^{mn} - 1 = \left(1 + \frac{0.048}{12}\right)^{12} - 1 = 0.04907$

Hence the best option is Option B with the highest effective rate

Week 4 Workshop Quiz (Financial Mathematics I) Time – 10 minutes

Module 3 – Financial Mathematics II

Brief Summary

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Annuities

- Annuities are a special case of multiple cash flow streams, where the cash flows are of equal size and occur at regular time intervals
- Examples include:
  - A constant superannuation payment made to retirees on a monthly basis throughout the remainder of their lives
  - Rentals paid in arrears
  - Interest payments on a bond

Annuities (cont)

Asset 1 : Cash flows received annually

Year	0	1	2	3	4	5	6	7	8
			$2.3	$2.3	$2.3	$2.3	$2.3	$2.3	

Asset 2 : Cash flows received every two years

Year	0	1	2	3	4	5	6	7	8
			$4		$4		$4		$4

Both of these cash flows are annuities

Annuities (cont)

- Types of annuity:
  - Ordinary annuity
    - Cash flows occur at the end of each period and, hence, the first cash flow occurs at the end of the first year
  - Annuity due
    - Cash flows occur at the beginning of each period and, hence, the first cash flow occurs immediately
  - Deferred annuity
    - The first cash flow in an annuity is delayed by x periods

Future value of an annuity

– To determine the accumulated value of an ordinary annuity of $PMT :

$$FV = PMT\left[\frac{(1+i)^n - 1}{i}\right]$$

where:

FV = the accumulated or future value of the annuity

PMT = the cash flow received/paid under the annuity

n = the number of cash flows that form the annuity

i = the compound interest rate per period

Present value of an annuity (cont)

– Alternatively, to determine the present value of an ordinary annuity of *$PMT*:

$$PV = PMT\left[\frac{1-(1+i)^{-n}}{i}\right]$$

where:

PV = the present value of the annuity

PMT = the cash flow received/paid under the annuity

n = the number of cash flows that form the annuity

i = the compound interest rate per period

Present value of a growing annuity

– To determine the present value of a growing ordinary annuity of *$PMT*:

$$PV = PMT \times \frac{1}{i-g}\left[1 - \frac{(1+g)^n}{(1+i)^n}\right]$$

where:

PV = the present value of the annuity

PMT = the cash flow received/paid under the annuity

n = the number of cash flows that form the annuity

i = the compound interest rate per period

g = the constant growth rate over the life of the annuity

Perpetuities

- Perpetuities are a special type of annuity where the cash flows are expected to go on forever
- The present value of a perpetuity:

$$PV = \frac{PMT}{i}$$

where:

PV = the present value of the perpetuity

PMT = the cash flow received/paid under the perpetuity

i = the compound interest rate per period

Constant Growth Perpetuity

- Constant growth perpetuities are a special type of perpetuity where the cash flows are expected to grow at a constant rate forever
- The present value of a constant growth perpetuity:

$$PV_t = \frac{PMT_{t+1}}{i - g}$$

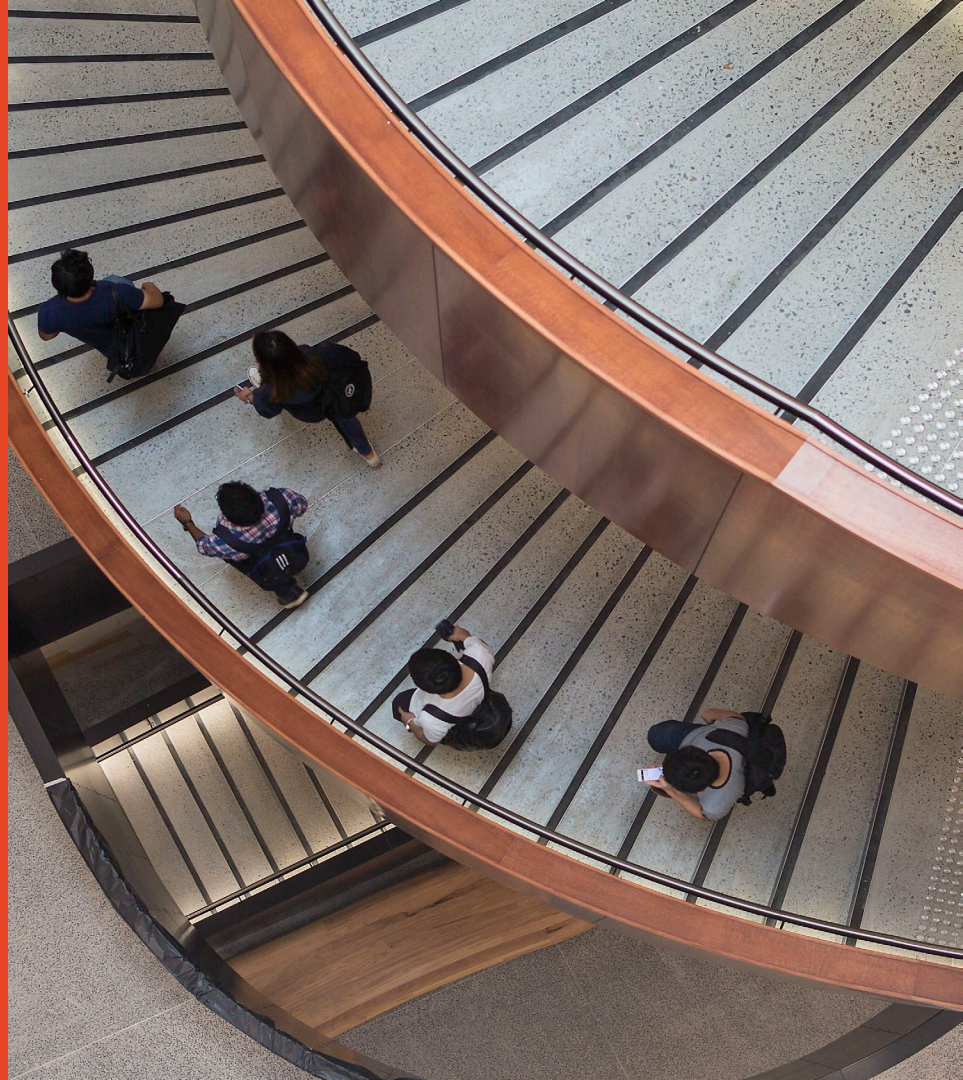
- Where

  g = the constant growth rate

Key Concept

- When using the annuity equation (including perpetuity equations) the compound frequency on the interest rate MUST MATCH the cash flow interval

Workshop Questions



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Question One

Oliver Queen is considering investing in Project Arrow. The project is expected to produce cash flows of $1m every three years with the first cash flow in 1½ years' time. The project is expected to produce a total of 15 cash flows over its life. The appropriate discount rate is 8.5% pa compounded monthly.

a) *What initial cost would result in Mr Queen being indifferent about whether to undertake Project Arrow?*

b) *Another project Mr Queen is considering, Project Hood, has the same present value as Project Arrow but produces a cash flow every month for 5 years with the first cash flow coming in one month's time. How much is the monthly cash flow from Project Hood?*

Question Two

a) *You are interested in buying a property that costs $500,000. You have saved up $62,000 for a deposit. Bank East Ltd is willing to lend you the balance at 4.8% pa compounded daily over a 30-year term. How much will your monthly repayments be?*

b) *For the property in the previous question, suppose you are required to pay a 3.6% stamp duty (tax) on the purchase. If the stamp duty is deducted from your deposit how much would your monthly repayments increase by?*

Question Three

Today is your 30th birthday. Given the recent changes in the tax law relating to retirement, the earliest you can retire is your 60th birthday. Your employer contributions to your superannuation are $2,500 p.a. on each birthday, starting immediately and continuing up to and including your 60th birthday. You estimate that you will need $60,000 per annum to live on from your 61st birthday to your 90th birthday (inclusive). You estimate that you can earn 12% p.a. between now and your 90th birthday.

a) *Can you afford to retire on your 60th birthday? Demonstrate your answer with calculations.*

b) *Given your savings plan, how much can you expect to receive as income each year when you retire – that is, from your 61st to your 90th birthday?*

Question Four

- *Suppose in question 3, we decided to only spend our interest income during retirement rather than consume any principle. How much lower would our retirement income be than the answer to part b?*

Question Five

- *Our company is considering the purchase of a new machines. Both perform the same function. Machine Alpha lasts for 5 years whilst Machine Beta lasts for 8 years. The required rate of return is 10% pa compounded annually. The annual costs of running each machine are given below ($000). Which machine should our company purchase?*

Year	0	1	2	3	4	5	6	7	8
Alpha	-20	-5	-5	-5	-5	-5			
Beta	-10	-8	-8	-8	-8	-6	-6	-6	-6

Question Six

Michael Brown is considering an investment that generates the annual cash flow stream given below. The appropriate require rate of return in 10% pa compounded annually. Demonstrate how Michael could determine its present value using the annuity equation.

Year	1	2	3	4	5	6	7	8	9
CF($)	200	400	800	200	400	800	200	400	800

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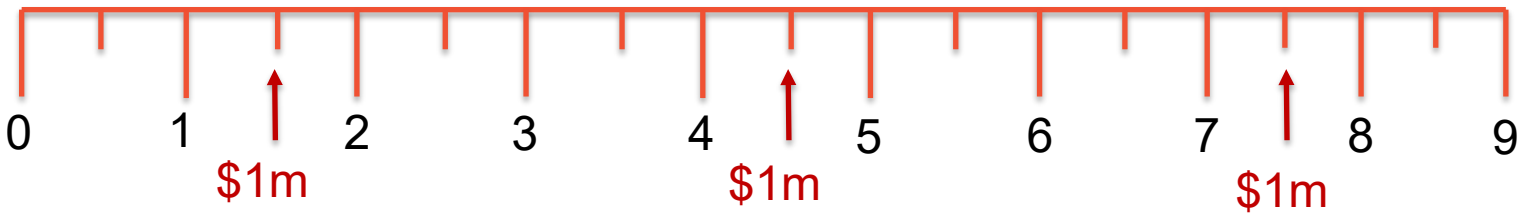

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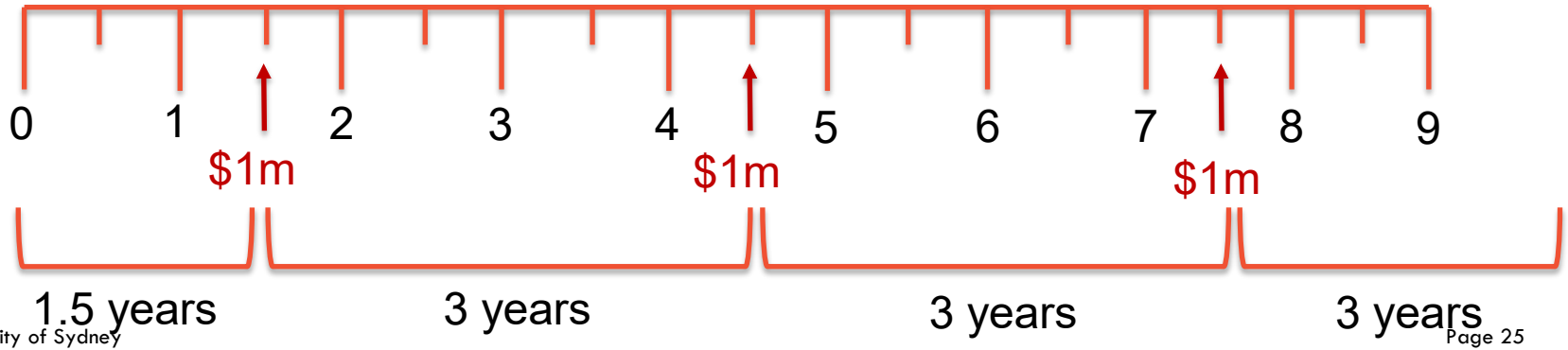

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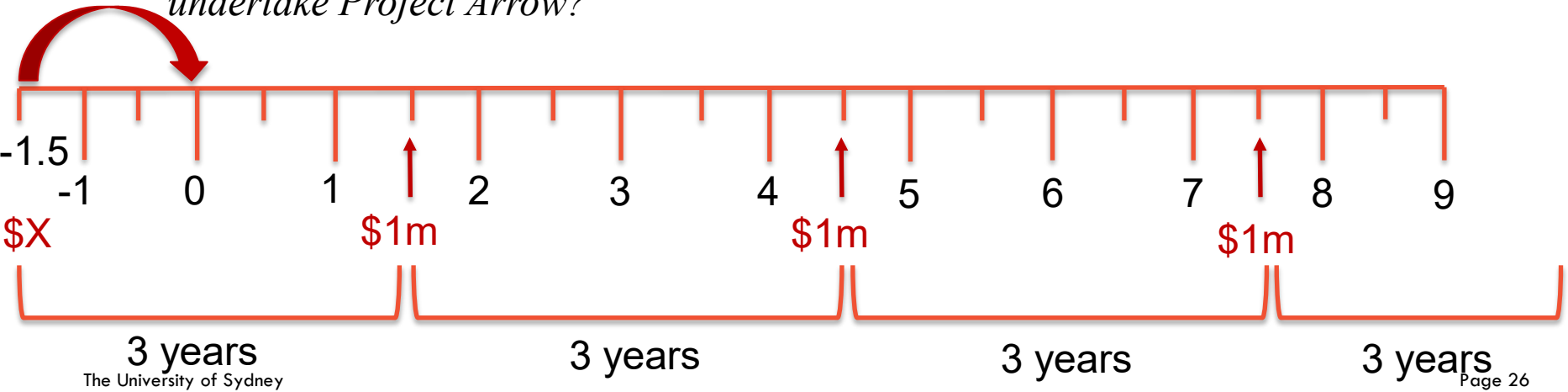

Question One

a) What initial cost would result in Mr Queen being indifferent about whether to undertake Project Arrow?

First determine the three-year effective rate:

$$Effective\ Rate = \left(1 + \frac{0.085}{12}\right)^{12x3} - 1 = 0.289$$

Then determine PV of ordinary annuity (at year -1.5):

Then determine PV at time zero:

$$PV_{-1.5} = \$1m\left[\frac{1-(1+0.289)^{-15}}{0.289}\right] = \$\ 3{,}380{,}152.30$$

This is the price at which he would be indifferent

$$PV_0 = \$\ 3{,}380{,}152.30\ \text{x}\left(1 + \frac{0.085}{12}\right)^{18} = \$3{,}838{,}076.48$$

Question One

b) Another project Mr Queen is considering, Project Hood, has the same present value as Project Arrow but produces a cash flow every month for 5 years with the first cash flow coming in one month's time. How much is the monthly cash flow from Project Hood?

This is an ordinary annuity calculation

Effective rate per month = 0.085/12 = 0.0071

$$\$3{,}838{,}076.48 = PMT\left[\frac{1-(1+0.0071)^{-60}}{0.0071}\right]$$

PMT= $78,744.02 per month

Question Two

a) You are interested in buying a property that costs $500,000. You have saved up $62,000 for a deposit. Bank East Ltd is willing to lend you the balance at 4.8% pa compounded daily over a 30-year term. How much will your monthly repayments be?

Question Two

a) You are interested in buying a property that costs $500,000. You have saved up $62,000 for a deposit. Bank East Ltd is willing to lend you the balance at 4.8% pa compounded daily over a 30-year term. How much will your monthly repayments be?

Effective rate per month = $(1+0.048/_{365})^{365*1/12} - 1 = 0.004007746$

Deposit = $62,000
Size of the loan = $500,000 - $62,000 = $438,000
Monthly repayment:

$$438000 = PMT\left[\frac{1-(1+0.004007746)^{-360}}{0.004007746}\right]$$

PMT = $2,300.50 per month.

Question Two

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Effective rate per month = $(1+0.048/_{365})^{365*1/12} - 1 = 0.004007746$

Stamp Duty = \$500,000 x 0.036 = \$18,000

Deposit (less stamp duty) = \$62,000 – \$18,000 = \$44,000

Size of the loan = \$500,000 - \$44,000 = \$456,000

Monthly repayment:

$$456000 = PMT\left[\frac{1-(1+0.004007746)^{-360}}{0.004007746}\right]$$

PMT = \$2,395.03 per month. Repayments went up \$94.52 per month due to the stamp duty

Question Three

Today is your 30th birthday. Given the recent changes in the tax law relating to retirement, the earliest you can retire is your 60th birthday. Your employer contributions to your superannuation are $2,500 p.a. on each birthday, starting immediately and continuing up to and including your 60th birthday. You estimate that you will need $60,000 per annum to live on from your 61st birthday to your 90th birthday (inclusive). You estimate that you can earn 12% p.a. between now and your 90th birthday.

a) *Can you afford to retire on your 60th birthday? Demonstrate your answer with calculations.*

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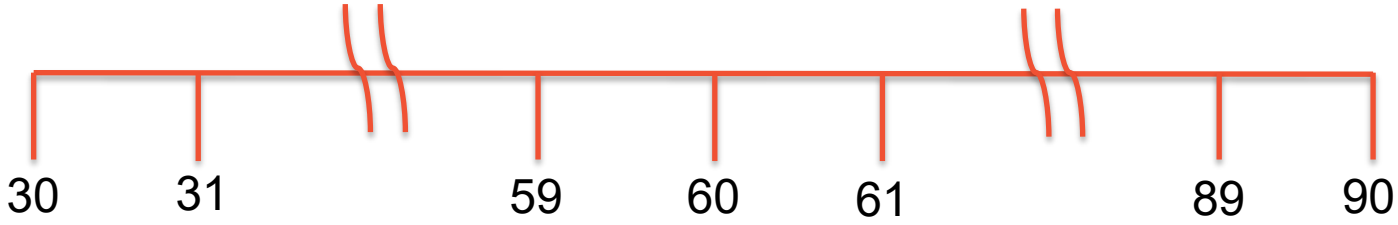

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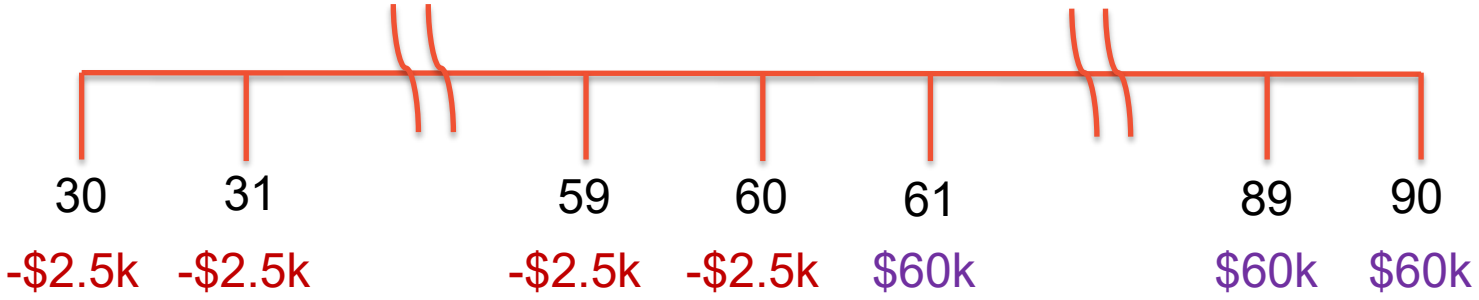

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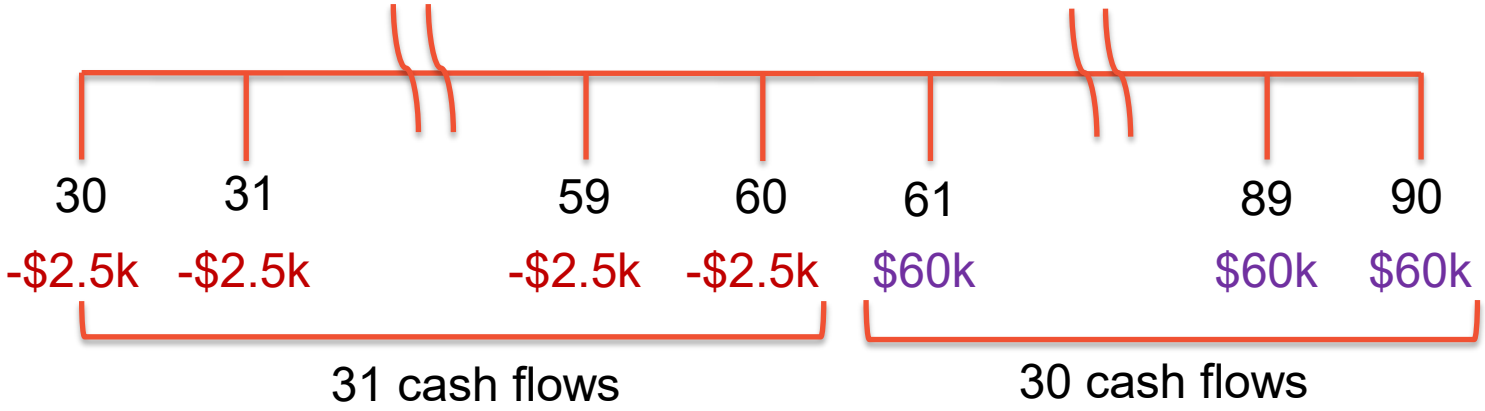

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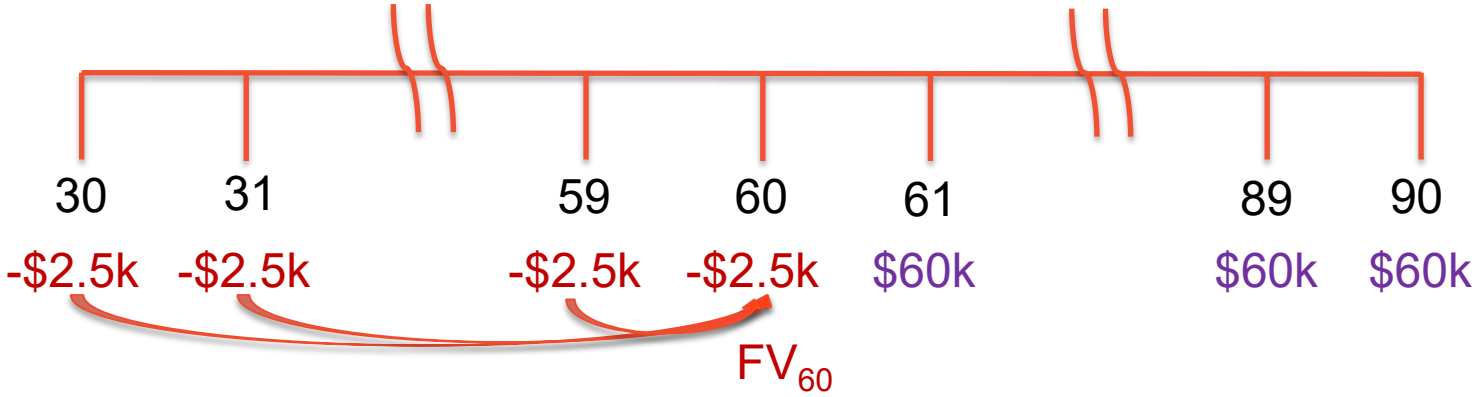


Question Three

a) Can you afford to retire on your 60th birthday? Demonstrate your answer with calculations.

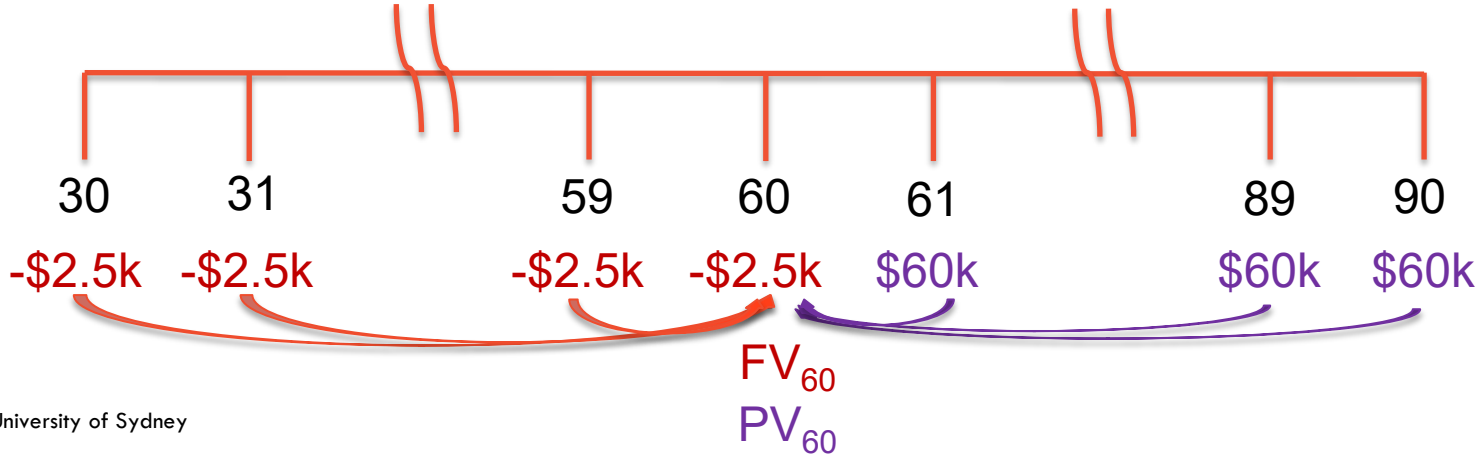
The first step is to calculate the amount of money that will have been accumulated by your 60th birthday. Given that there is a payment on your 60th birthday, and also a payment on your 30th birthday, this is a 31-year annuity ordinary annuity.

$$FV = PMT\left[\frac{(1+i)^n - 1}{i}\right]$$

$$FV_{60} = 2500\left[\frac{1.12^{31} - 1}{0.12}\right] = \$678{,}231.52$$

Question Three

Today is your 30th birthday. Given the recent changes in the tax law relating to retirement, the earliest you can retire is your 60th birthday. Your employer contributions to your superannuation are $2,500 p.a. on each birthday, starting immediately and continuing up to and including your 60th birthday. You estimate that you will need $60,000 per annum to live on from your 61st birthday to your 90th birthday (inclusive). You estimate that you can earn 12% p.a. between now and your 90th birthday.



Question Three

a) Can you afford to retire on your 60th birthday? Demonstrate your answer with calculations.

This amount is the present value of the 30-year annuity that comprises the annual withdrawals from your 61st birthday to your 90th birthday.

$$PV = PMT\left[\frac{1-(1+i)^{-n}}{i}\right]$$

$$PV_{60} = 60000\left[\frac{1-(1.12)^{-30}}{0.12}\right] = \$483{,}311.04$$

You would only require \$483,311 to be accumulated by your 60th birthday to be able to withdraw \$60,000 per year from your 61st birthday to your 90th birthday, so you can afford to retire on your 60th birthday.

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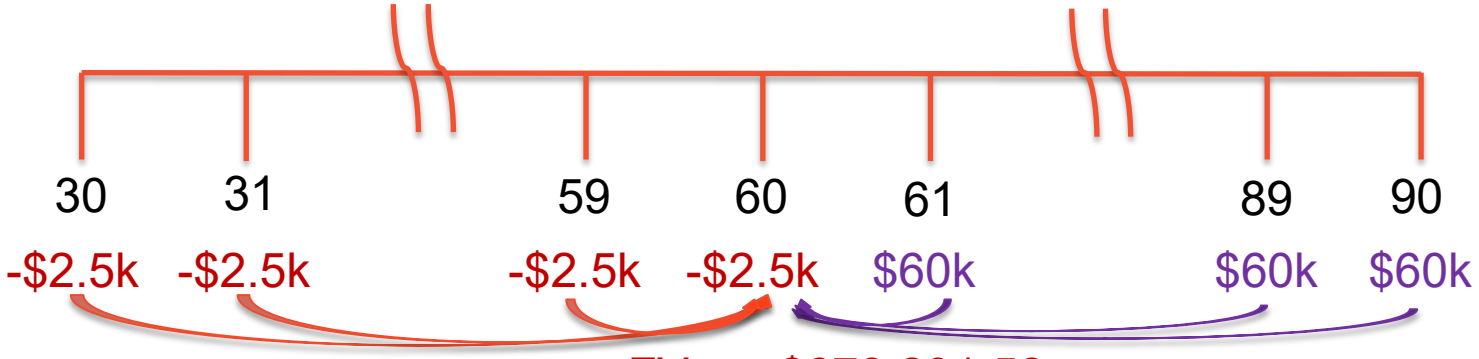

FV_{60} = $678,231.52

PV_{60} = $483.311.04

Question Three

b) Given your savings plan, how much can you expect to receive as income each year when you retire – that is, from your 61st to your 90th birthday?

Using the accumulated balance we now calculate the PV of the retirement annuity cash flows:

$$678{,}231.52 = PMT\left[\frac{1-(1.12)^{-30}}{0.12}\right]$$

$$PMT = \$84{,}198.14$$

This is the amount that can be withdrawn each year from your 61st birthday to your 90th birthday.

Question Four

- *Suppose in question 3, we decided to only spend our interest income during retirement rather than consume any principle. How much lower would our retirement income be than the answer to part b?*

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- *Suppose in question 3, we decided to only spend our interest income during retirement rather than consume any principle. How much lower would our retirement income be than the answer to part b?*

- With an accumulated account balance of $678,231.52 the amount of interest we would earn each year is 0.12 x $678,231.52 = $81,387.78

- Thus, we would need to sacrifice only $2810.36 ($84,198.14 - $81,387.78) per year to preserve the entire balance of $678,231.52

Question Five

– *Our company is considering the purchase of a new machines. Both perform the same function. Machine Alpha lasts for 5 years whilst Machine Beta lasts for 8 years. The required rate of return is 10% pa compounded annually. The annual costs of running each machine are given below ($000). Which machine should our company purchase?*

Year	0	1	2	3	4	5	6	7	8
Alpha	-20	-5	-5	-5	-5	-5			
Beta	-10	-8	-8	-8	-8	-6	-6	-6	-6

Question Five

Firstly, find the present value of the two cash flow streams

Machine Alpha

$$PV = -20{,}000 - 5{,}000\left[\frac{1-(1+0.10)^{-5}}{0.10}\right] = \text{-\$38,953.93}$$

Machine Beta

$$PV = -10{,}000 - 8{,}000\left[\frac{1-(1+0.10)^{-4}}{0.10}\right] - \frac{6{,}000\left[\frac{1-(1+0.10)^{-4}}{0.10}\right]}{(1.10)^4} = \text{-\$48,349.29}$$

Question Five

However, since the projects are of different lengths we then need to find the equivalent annuity so that we can compare them. This involves find the PMT for annuities with the present values given above

Machine Alpha

$$38{,}953.93 = PMT\left[\frac{1-(1+0.10)^{-5}}{0.10}\right]$$ therefore PMT = $10,275.95

Machine Beta

$$48{,}349.29 = PMT\left[\frac{1-(1+0.10)^{-8}}{0.10}\right]$$ therefore PMT = $9,062.79

Therefore, machine Beta is the best option

Question Six

Michael Brown is considering an investment that generates the annual cash flow stream given below. The appropriate require rate of return in 10% pa compounded annually. Demonstrate how Michael could determine its present value using the annuity equation.

Year	1	2	3	4	5	6	7	8	9
CF($)	200	400	800	200	400	800	200	400	800

Question Six

Here we have in effect three annuities. This can solved using various combinations of the original cash flows but I have chosen to split up the streams in as given in the table below. I have labelled the streams A, B and C.

Year	1	2	3	4	5	6	7	8	9
A	200	200	200	200	200	200	200	200	200
B		200			200			200	
C			600			600			600

We can then find the present value of each of these cash flow streams using the annuity equation.

For stream A we have an ordinary annuity of \$200 per year for 9 years and hence its present value will be:

$$PV_A = 200\left[\frac{1-(1+0.10)^{-9}}{0.10}\right] = \$1151.80$$

Question Six

For stream B we have an ordinary annuity of $200 with an interval of three years (although the first period is only one year). To solve this we need an effective three year interest rate. This rate is calculated as:

$$Effective\ Rate = \left(1 + \frac{0.10}{1}\right)^{1x3} - 1 = 0.331$$

The present value of the annuity can then be found using EITHER of the methods below:

$$PV_B = \left(200\left[\frac{1-(1+0.331)^{-3}}{0.331}\right]\right)(1.10)^1 = \$382.77$$

OR

$$PV_B = \frac{\left(200+200\left[\frac{1-(1.331)^{-2}}{0.331}\right]\right)}{(1.10)^2} = \$382.77$$

Question Six

For stream C we have an ordinary annuity of \$600 with an interval of three years. To solve this we once again use the effective three year interest rate. The present value of the annuity can be found as:

$$PV_C = 600\left[\frac{1-(1+0.331)^{-3}}{0.331}\right] = \$1043.93$$

The total value of the cash flow stream is aggregate of the present values of streams A, B and C. Thus, the present value is \$1151.80 + \$382.77 + \$1043.93 = \$2578.50

The End

Next topic we look valuation of debt and equity securities